



CSC1107: Domain Applications 2

Module Co-ordinator: Dr. Ellen Rushe

Project Description

Note: This assignment is worth 95% of your overall grade for the module. **Please follow instructions carefully and note all relevant deadlines.**

Section	Deadline
1. Problem Outline	3rd October @ 17:00
2. Code & Report Resources	30th November @ 17:00
3. Final Report	28th November @ 17:00
4. Presentation	In-class November 24th, 15:00-17:00
5. Interview	In-class November 24th, 17:00-18:00

0 Introduction

In this assignment, you will be required to use the insights that you have gained from the presentations in this module in order to create a project that solves a real-world problem in an interesting and insightful way. This assignment has three sections:

1. **Problem Outline:** In this document, you will briefly outline a real-world problem you want to solve or an overarching question you would like to answer with data analysis and/or machine learning. You will also be required to outline the relevant tasks/research questions, choose a dataset or set of datasets to address the tasks outlined, assess the legal details of using your chosen data and consider the ethical implications of you chosen tasks.
2. **Code:** Using your problem outline and chosen dataset(s), perform appropriate analysis and/or train machine learning models using Python or R (or both) to address the tasks or questions described.
3. **Research Paper:** Using all of the above, write a research paper outlining the motivation for the problem, background research, dataset, implementation, experimental results, conclusions and ethical considerations. 3.
4. **Presentation:** Here, after finishing your project, you will present it in class or by creating a video.

This project has a total of **95 marks**. **Every calendar day late will incur a 10% penalty for each section that has a submission deadline**, except in the case of extenuating circumstances as outlined in the university policy.

1 Problem Outline (10 marks)

To complete this task, outline your project proposal, dataset, the dataset licence, GDPR details and ethical considerations following the template provided on Loop:

project_outline_template.doc

For an example project outline document see

project_outline_example.pdf

Submission Instructions: Upload as PDF with a file name in the following format:

<FirstName>_LastName_<student_id>.pdf
e.g. ellen_rushe_12345678.pdf

Deadline: 3rd October @ 17:00

1.1 Research Aims

Outline a real-world problem that you would like to solve. This should take the form of a problem title and **three research questions** that you would like to answer or tasks you would like to complete.

1.2 Dataset

Choose data from a **reputable source**, and outline the following information:

- The dataset(s)' source(s).
- Licence(s) assigned to dataset.
- Terms of licence.
- Whether the dataset can be used for non-commercial purposes (i.e. research purposes).

1.3 GDPR

In this section, you'll be required to consider your legal requirements under GDPR, including the legal basis for using data and any privacy measures you have put in place if you are processing any special category data.

1.4 Ethical Considerations

Here you will be required to give an outline of ethical considerations that should be made when using this information. This goes *beyond* simply discussing your legal responsibility. Instead, this should indicate any individual or societal harms that could arise from using the data that you have specified. Regardless of the data used here, **it is not acceptable to leave this section blank**. All data usage, analysis and models can have ethical implications.

2 Code (30 marks)

Here you should include all code used for your analysis and modeling. Code should be in either Python or R. You **must** cite any places where you used a significant block of code online using a comment over the block.

Comments should feature throughout all code files and markdown cells used where a significant block of explanation is necessary. Ensure tidy and easily understandable code, as this will be evaluated.

- All code files for your implementation/analysis placed in a folder named code.
- **Don't submit your data**. Provide a link to it in your report from Section 1.

Example zip file:

```
ellen_rushe_123456789.zip
|_ code
|_   amazing_analyis_notebook.ipynb
|_   amazing_analysis_notebook.html
|_   some_python_script.py
|_   .
|_   .
|_   .
|_   some_bash_script.sh
```

Submission Instructions: Upload as zip file to Loop.

Deadline: November 30th @ 17:00

3 Report (40 marks)

A template for the project report is available on Loop along with a marking rubric which sets out a clear set of guidelines for the expected content of the report. You should adhere to this in order to obtain a good mark. The use of Generative AI (e.g. ChatGPT) for any part of this project is **explicitly forbidden**. Part of this project is to develop your own writing and communication skills, so please take this opportunity to develop these skills independently.

Final submission: November 28th, 17:00

4 Presentation (15 marks)

Regardless of the presentation format, you should include the following information.

- Problem Description
- The solutions to this problem that are out there already, if any.
- Brief description of dataset
- Proposed Approach
- Result of Analysis

In-person

The presentation should be roughly 5 minutes long. There will be an additional 3 minutes for questions afterwards.

Video

The video should be roughly 5 minutes long. The video does not have to use slides, but must still include the same information as an in-class presentation. Feel free to be as creative as you like here!

5 Interview

If you choose to create a video for the presentation section of this project, you will then be required to have a 3 minute interview (not in front of the class) to discuss your project.

Submission Instructions: Upload as a PDF file or video to Loop.

Deadline: Before presentation session on November 24th @ 15:00